

IT24101148

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PS_LAB_8

```
1 setwd("C:\\Users\\UsEr\\OneDrive\\Desktop\\PS\\IT24101148_PS_Lab_8")
2 data <- read.table("Exercise - LaptopsWeights.txt", header = TRUE)
3 fix(data)
4 attach(data)
5 |
6
7 pop_mean<-mean(weight.kg.)
8 pop_sd<-sd(weight.kg.)
9
10 cat("Population Mean (True Mean):",pop_mean,"\n")
11 cat("Population Standard Deviation (True SD):",pop_sd,"\n")
12
13 #2
14 samples<-c()
15 sample_names <- c()
16
17 for (i in 1:25){
18   s <- sample(weight.kg.,6,replace =TRUE)
19   samples <- cbind(samples,s)
20   sample_names<-c(sample_names, paste('s',i))
21 }
22
23 colnames(samples)= sample_names
24 s.means<-apply(samples,2,mean)
25 s.sds<-apply(samples,2,sd)
26
27 #3
28
29 samplemean<-mean(s.means)
30 samplesd<-sd(s.means)
31
32 pop_mean
33 pop_sd
34
35 truesd=pop_sd/6
36 samplesd
37
```

```
> data <- read.table( Exercise - LaptopsWeights.txt , header = TRUE)
> fix(data)
> attach(data)
> pop_mean<-mean(weight.kg.)
> pop_sd<-sd(weight.kg.)
> cat("Population Mean (True Mean):",pop_mean,"\n")
Population Mean (True Mean): 2.468
> cat("Population Standard Deviation (True SD):",pop_sd,"\n")
Population Standard Deviation (True SD): 0.2561069
> #2
> samples<-c()
> sample_names <- c()
> for (i in 1:25){
+   s <- sample(weight.kg.,6,replace =TRUE)
+   samples <- cbind(samples,s)
+   sample_names<-c(sample_names, paste('s',i))
+ }
> colnames(samples)= sample_names
> s.means<-apply(samples,2,mean)
> s.sds<-apply(samples,2,sd)
> samplemean<-mean(s.means)
> samplesd<-sd(s.means)
> pop_mean
[1] 2.468
> pop_sd
[1] 0.2561069
> truesd=pop_sd/6
> samplesd
[1] 0.1149369
> |
```